

SEYMOUR JOHNSON AFB, NC, USA

WMO: 723066

Lat: 35.344N

Lon: 77.965W

Elev: 109

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13713

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	25.4	3.2	6.5	27.9	8.8	8.6	32.8	24.2	53.2	20.7	53.7	5.7	280	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	96.4	76.2	93.6	76.1	91.4	75.3	80.6	87.8	79.4	86.9	78.4	85.9	9.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.0	82.5	77.7	144.6	81.7	76.7	139.5	81.1	44.2	88.1	42.9	86.8	41.8	86.1	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	17.2	15.4	14.0	99.8	5.3	2.6	10.2	101.7	7.1	103.2	4.1	104.6	0.2	106.5
			12.5	82.6	5.0	2.4	8.9	84.3	6.0	85.7	3.2	87.1	-0.4	88.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	43.9	48.0	53.7	63.0	71.0	78.5	81.4	79.6	74.2	64.2	53.4	47.4
	DBStd	15.26	10.94	10.18	9.43	7.91	6.83	5.28	3.93	4.15	5.42	7.76	8.53	9.62
	HDD50	706	254	145	71	5	0	0	0	0	0	3	60	168
	HDD65	2687	658	482	367	131	27	1	0	0	3	110	357	551
	CDD50	5546	63	89	185	394	652	854	972	918	727	444	162	86
	CDD65	2055	3	6	17	70	214	404	507	453	280	86	10	5
	CDH74	19094	11	41	161	643	1912	4023	5189	4250	2110	666	62	26
	CDH80	7974	0	3	29	156	706	1853	2435	1856	771	161	2	2
Wind	WSAvg	6.8	7.1	7.7	8.1	8.1	7.2	6.7	6.3	5.8	6.2	6.1	6.2	6.6
Precipitation	PrecAvg	48.5	4.3	3.5	4.2	3.3	3.7	4.0	5.8	5.5	5.1	2.8	2.9	3.1
	PrecMax	70.9	8.1	7.2	7.6	8.2	9.4	10.9	13.6	16.8	30.5	9.6	6.6	7.4
	PrecMin	34.6	1.1	0.7	0.7	0.5	0.7	0.9	1.1	1.3	0.1	0.0	0.5	0.6
	PrecStd	7.4	1.8	1.7	1.7	1.8	1.8	2.2	3.0	3.0	5.0	2.2	1.6	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	79.1	83.6	87.5	95.0	98.9	99.8	98.3	93.9	89.1	79.3	77.4
		MCWB	66.1	64.5	63.3	67.4	73.1	75.9	76.9	76.9	73.0	71.9	66.3	64.3
	2%	DB	70.0	73.7	78.6	84.2	90.6	95.5	96.2	94.6	90.7	84.1	75.4	72.0
		MCWB	62.4	61.4	63.1	66.3	71.5	75.9	77.3	76.4	73.0	70.2	63.7	64.2
	5%	DB	65.8	69.2	74.3	81.5	87.5	92.7	93.4	91.6	87.9	81.0	72.0	67.9
		MCWB	59.9	59.1	60.9	65.3	70.6	75.1	77.4	76.6	73.0	68.0	62.9	62.7
	10%	DB	61.4	64.9	70.1	77.6	83.9	89.9	91.0	89.2	84.9	77.4	68.1	63.6
		MCWB	55.9	57.2	59.0	64.1	69.7	74.5	76.6	75.5	72.4	67.3	61.5	59.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.6	67.8	69.9	74.9	76.8	80.4	83.1	82.1	79.7	76.9	71.1	69.2
		MCDB	71.9	73.9	75.2	80.3	84.1	89.7	89.2	88.2	86.0	81.6	74.1	73.0
	2%	WB	64.8	65.3	66.9	70.8	75.0	78.5	81.1	80.3	78.0	74.9	68.4	66.1
		MCDB	68.6	71.4	72.5	77.0	83.4	87.3	88.3	86.9	83.8	79.5	71.4	69.9
	5%	WB	61.5	62.2	64.6	68.9	73.4	77.5	79.8	79.1	76.6	72.8	65.9	63.0
		MCDB	65.0	67.1	70.0	75.7	81.6	86.6	87.6	86.1	82.2	77.2	70.1	66.4
	10%	WB	56.8	58.3	62.2	66.7	72.2	76.5	78.6	78.0	75.2	70.1	62.9	59.5
		MCDB	60.2	64.0	68.6	74.3	79.9	85.3	86.5	85.1	80.3	74.7	66.6	63.3
Mean Daily Temperature Range	5% DB	MDBR	18.8	20.8	21.6	22.1	20.7	19.5	18.4	18.0	17.7	21.2	21.3	18.7
		MCDBR	24.2	25.0	26.8	25.5	23.9	22.4	21.2	20.6	21.4	23.5	24.6	23.0
		MCWBR	17.7	16.0	14.0	11.6	8.8	7.3	6.9	6.8	7.9	10.8	14.6	15.9
	5% WB	MCDBR	20.3	23.1	21.4	20.5	19.9	19.9	19.0	17.8	16.5	17.4	19.6	21.1
		MCWBR	17.8	17.7	15.2	12.3	8.7	7.6	7.7	7.6	7.8	10.2	14.9	17.2
Clear-Sky Solar Irradiance	taub	0.316	0.332	0.361	0.405	0.458	0.503	0.537	0.524	0.423	0.371	0.334	0.317	
	taud	2.494	2.442	2.394	2.272	2.163	2.083	2.002	2.030	2.313	2.433	2.493	2.526	
	Ebn at Noon	282	289	288	279	265	252	242	243	265	271	272	274	
	Edh at Noon	26	31	36	42	48	52	56	53	38	31	26	24	
All-Sky Solar Radiation	RadAvg	829	1047	1380	1741	1916	2027	1932	1741	1452	1188	912	728	
	RadStd	54	113	95	95	115	142	125	112	95	110	149	71	58

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+290	+203

Nomenclature: See separate page